
Exercises: Advanced Lesson 5

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Your Math Class: MATH 7013 – Advanced Mathematical Modeling

Note: Some of the following exercises are expressed in every day mathematical notation rather than proper *Mathematica* notation. You MUST convert every day notation into *Mathematica* notation for your code to work properly. For example:

Equations must be changed from "=" to "==" to work correctly (i.e. $x^2 == 2x$ not $x^2 = 2x$).

Functions must be expressed as $f[x]$ and not $f(x)$ to work correctly.

Functions must be defined using "_" to work correctly (i.e. $f[x_] = 2x^2 + 1$ and not $f[x] = 2x^2 + 1$).

Mathematica functions must start with a capital (i.e. $\text{Sin}[x]$ not $\text{sin}[x]$).

Your functions should start with a lower case letter (i.e. $f[x]$ not $F[x]$).

Special symbols must be entered correctly (i.e. e from the palette and not e).

These exercises begin your training in making this conversion automatically.

Remember to clear all variables that you will use below: `Clear[ ... ]`

```
Clear[a, x, i, j, list3, list4, mat5, courseNumbers, courseNames, me];
```

1: Type in a list that contains your name (as a string), your age (as an integer) and your major (as symbols i.e. variables). Evaluate this expression.

```
me = {"Mikey Ferguson", 27, mathematics}
```

2: Type in a list of lists which contains a list of the course numbers that you are taking this semester (as integers), a list of the corresponding course names (as symbols i.e. variables). Display this list using: `TableForm`. Evaluate this expression.

```
courseNumbers = {7013, 7023, 7113};
courseNames = {advancedMathematicalModeling, realAnalysis, numericalAnalysis};
TableForm[{courseNumbers, courseNames}]
```

3: Generate a list of integers from 0 to 20 using the Table command.

```
list3 = Table[i, {i, 0, 20}]
```

4: Generate a list of EVEN integers from 0 to 20 using the Table command.

```
list4 = Table[i, {i, 0, 20, 2}]
```

5: Part 1: Generate a list of lists using the Table command which represents a 4x5 matrix containing the elements $\sin[i] \cos[j]$.

Part 2: Display this list of lists using: MatrixForm. Evaluate this expression.

```
mat5 = Table[Sin[i] Cos[j], {i, 1, 4}, {j, 1, 5}];
MatrixForm[mat5]
```

6: Sort the list containing your course numbers. Sort the list containing the corresponding course names.

```
{Sort[courseNumbers], Sort[courseNames]}
```

7: Reverse the list generated in Exercise 3.

```
Reverse[list3]
```

8: Drop the first three elements from the list generated in Exercise 3. Drop the last three elements from the list generated in Exercise 3.

```
{Drop[list3, 3], Drop[list3, -3]}
```

9: Take the first three elements from the list generated in Exercise 3. Take the last three elements from the list generated in Exercise 3. Take elements 3, 4, and 11 from the list generated in Exercise 3.

```
{Take[list3, 3], Take[list3, -3], list3[[3, 4, 11]]}
```

10: Insert a second element "Fun and games" into the list of your course names in Exercise 2.

```
Insert[courseNames, "Fun and games", 2]
```

11: Replace the second course name in the list of your course names in Exercise 2 with "Fun and games".

```
ReplacePart[courseNames, 2 → "Fun and games"]
```

12: Determine the length of the list in Exercise 4:

```
Length[list4]
```

13: Determine the dimensions of the list of lists in Exercise 5:

```
Dimensions[mat5]
```

14: Part 1: Take the intersection of the lists in exercises 3 and 4.

Part 2: Display this list using: ColumnForm. Evaluate this expression.

```
ColumnForm[list3 ∩ list4]
```

15: Join the lists in exercises 3 and 4. Display this list using: ColumnForm. Evaluate this expression.

```
ColumnForm[Join[list3, list4]]
```

16: Partition the list in Exercise 3 to contain lists of length 4. Display this list of lists using: TableForm. Evaluate this expression.

```
list16 = Partition[list3, 4];  
TableForm[list16]
```

17: Partition the list in Exercise 3 to contain lists of length three which each contain lists of length 2. Display this list of lists of lists using: MatrixForm. Evaluate this expression.

```
MatrixForm[Partition[Partition[list3, 2], 3]]
```

18: Using the `[[...]]` notation, extract the third element from the list in Exercise 3 and the second element from the list of lists in Exercise 16.

```
{list3[[3]], list16[[2]]}
```

19: Using the `[[...]]` notation, extract the element corresponding to the second row and third column from the list of lists in Exercise 5.

```
mat5[[2, 3]]
```

20: Using the `[[...]]` notation, extract the second row from the list of lists in Exercise 5.

```
mat5[[2]]
```

21: It is possible to use JUST the `[[...]]` notation to extract the second column from the list of lists in Exercise 5. You may have to look up the Part command in Help.

```
mat5[[All, 2]]
```

22: Evaluate directly: $e^{\{1,2,a,x,\text{"string"},\pi\}}$

```
 $e^{\{1,2,a,x,\text{"string"},\pi\}}$ 
```

23: Evaluate: **Log[{1, 2, a, x, "string", π}]** indirectly by using the Map command. Next, directly obtain the list of approximate real numerical values for the previous result using the N Command.

```
logs = Log /@ {1, 2, a, x, "string", π}  
N[logs]
```

24: Apply Plus to the list in Exercise 4.

```
Plus @@ list4
```

25: Flatten out the list of lists in Exercise 5.

```
Flatten[mat5]
```